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An EEG Investigation of Error-Related Negativity in Human-Robot Interactions

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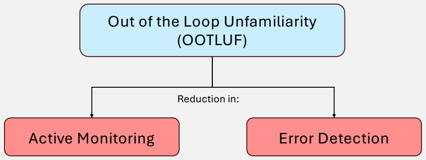
Introduction

- Assistive robot operators can minimize worker injuries and maximize efficiency in manufacturing settings (Kaasinen et al., 2020), but "**noise**" may prevent a robot from performing as desired.
- Effective **human robot interaction** (HRI), requires human **calibrated trust** to effectively monitor a robot's performance (de Visser, et al., 2019)
- As the level of automation increases, humans may become less familiar with system behavior known as **Out of the Loop Unfamiliarity (OOTLUF)** (Wickens, et al., 2015).



- **Error-Related Negativity (ERN**) is an EEG marker elicited when a detected action deviates from the expected action, appearing 50-100 *ms* after an error is observed (Falkenstein et at., 1991)
- **We hypothesize that when the robot performs unreliably, a higher ERN amplitude will be observed.**

Limitations & Future Steps

- Sample size
- Learning effect
- Add random order of unreliable trials

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Methods

- 23 participants co-lifted a rectangular object with a UR10e robotic arm through a 5-segment obstacle course, completing 16 trials and trust in automation on the Jian et al., (2000) trust scale.
  - 8 reliable (speed 0.2 *m/s*)
  - 8 unreliable (speed 0.1 *m/s*, 0.2 *m/s*, 0.4 *m/s*)

Figure 1: Experimental setup of co-lifting task with EEG cap

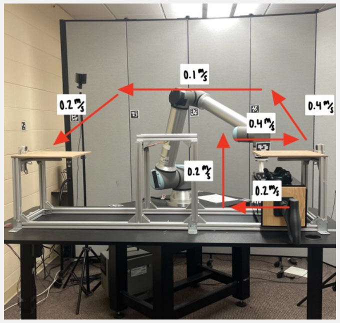


Figure 2: Unreliable robot condition path with varying velocities

- EEG was recorded at 500 Hz using a 32-electrode cap using the 10-20 system and preprocessed to remove artifacts and noise, specifically looking at Fz and Cz.

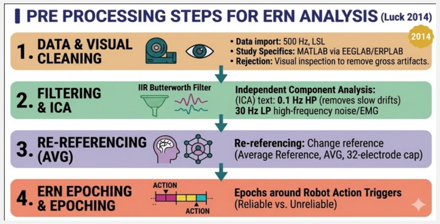


Figure 3: Preprocessing steps taken before ERN analysis RStudio

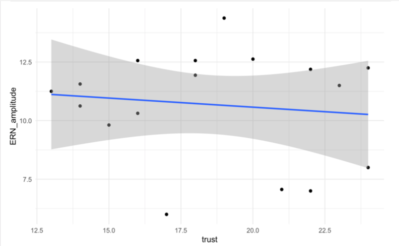
- A **two-tailed paired t-test** was used to measure the ERN amplitude during both conditions.
- **Pearson product-moment correlation tests** and **linear regression** were used to see if ERN amplitude was affected by trust propensity questionnaire results.

Results

- No significant difference was observed in ERN peak amplitude across reliable and unreliable robot trials (p_{Cz} = .2866, p_{Fz} = .9005) for electrodes Fz and Cz

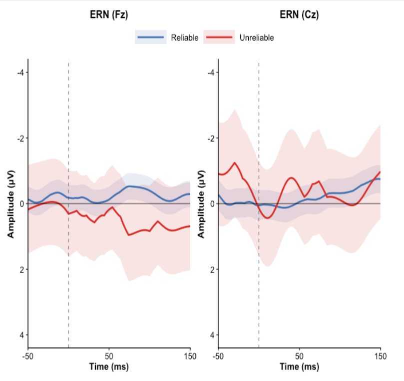
Electrode	n	Reliable Mean (Hz)	Unreliable Mean (Hz)
Fz	23	0.120+=3.14	-0.144+=7.31
Cz	19	1.380+=3.78	-0.923+=10.23

Table 1: ERN amplitude mean and standard deviation for both conditions and for each electrode



Graph 1: Linear Regression of trust in automation scores and ERN peak amplitude

- No evidence that pre-experiment trust in robots predicts ERN peak amplitude (p = .6473 , R^2 = 0.01432, F = .2719).



Graph 2: ERN Amplitude graph for both electrodes comparing unreliable and reliable conditions

Discussion

- No evidence found that an unreliable robot elicits neural activity consistent with the expected ERN effect. Participants may become "out of the loop", which decreased their active monitoring.
- In manufacturing settings, workers may not be active monitoring, making them unprepared for errors done by the robot. This could result in worker injuries or system failures (de Visser, et al. 2019)
- An expected, larger, mean ERN amplitude at the Cz site could indicate that ERN would appear with increased statistical power, as Cz has been found to elicit a larger ERN amplitude (Konno & Izawa, 2025)